# Free-fall lab

**antocciphysics**

**Name:** _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

**LEARNING TARGET** Estimate g from measured motion and discuss measurement limits.

**SUCCESS CRITERIA - I can...**

1. Keep supplied examples separate from measurements.
2. Find acceleration from a velocity-time slope.
3. Use repeated trials and explain uncertainty.

**Core:** teacher selects from 1-7. **Stretch:** 8 after checking your work.

**MARZANO SELF-CHECK**

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

## 01 Warm-up: bring back an earlier idea

Two equal 3000 kg masses are 5 m apart. Use $G=6.67\times10^{-11}$ N·m²/kg². Find force. By what factor is an answer wrong if r rather than r² is used?

**Marzano check - circle:** ① ② ③

## 02 Worked example: explain the model

**Reference:** $a = \Delta v/\Delta t$; $v_f = v_i + a\,\Delta t$.
Velocity-time slope estimates acceleration. Downward is positive. Use the approved apparatus; distinguish supplied from measured data.

**Worked reasoning:** Supplied trial values 8, 10, 12 m/s² have mean 10 m/s² and range 8-12. The spread describes variation; averaging does not remove every systematic error.

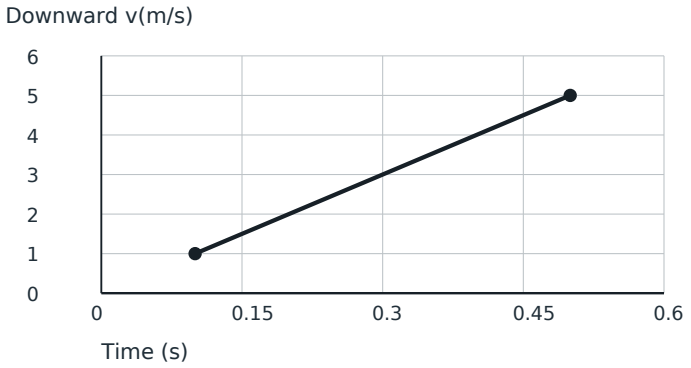
**Your turn:** Plot the three values on a number line and mark the mean. Name an error that could shift all trials together.

| Trial | Estimated g (m/s²) |
|---|---|
| 1 | 8 |
| 2 | 10 |
| 3 | 12 |

**Marzano check - circle:** ① ② ③

## 03 Read and interpret the model

Use the supplied velocity-time graph to estimate acceleration. Show the change in velocity and elapsed time.




**Marzano check - circle:** ① ② ③

## 04 Apply the physics

Three video trials estimate g as 9.6, 9.9 and 10.2 m/s². Find the mean and half-range. Compare the mean with 9.8 m/s² using percent difference from that reference. Label a sketch of the measurement interval.

| Trial | Estimated g (m/s²) |
|---|---|
| 1 | 9.6 |
| 2 | 9.9 |
| 3 | 10.2 |

**Marzano check - circle:** ① ② ③

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Continue down each column.

## 05 Explain your reasoning

Plan three repeated trials. Name a controlled condition, a measurement you will record and a reason not to include a badly pushed release without comment.

**Marzano check - circle:** ① ② ③

## 06 Review: connect an earlier idea

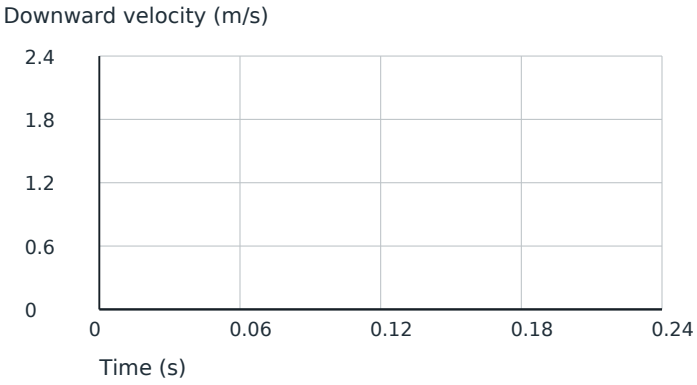
A 4 kg lamp hangs at rest from a vertical rope. Use g=9.8. Find tension. Then find tension if it accelerates upward at 3 m/s².
Draw the forces on the chosen object and label their sources.

Add and label force arrows.

**Marzano check - circle:** ① ② ③

## 07 Exit check

A video gives downward velocities of 0.15 m/s at 0.02 s and 2.11 m/s at 0.22 s. Plot the points and find the slope. Explain why fitting several reliable measurements can be better than using one very short interval.




**Marzano check - circle:** ① ② ③

## 08 STRETCH: test the model

Supplied estimates are 9.4, 9.8, 10.2 m/s². Find mean, half-range and percent deviation of mean from 9.81. Could zero spread guarantee accuracy?

| Group / trial | Values (m/s²) | Mean | Spread |
|---|---|---|---|
| 1 | ____ | ____ | ____ |
| 2 | ____ | ____ | ____ |

**Marzano check - circle:** ① ② ③